

Cambridge (CIE) IGCSE Computer Science Paper 2: Algorithms, Programming and Logic (Set B)

Tuesday 6 May 2025

Morning (Time: 1 hour 45 minutes)

Total marks

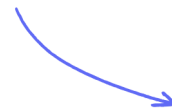
/ 76

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.
- Calculators must not be used in this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 (a) State five file-handling operations from the list below:

- calculate
- close
- count
- create
- input
- open
- output
- print
- read
- sort
- search
- test
- total
- write

(5 marks)

(b) Explain why a program might need to store data in a file

(3 marks)

2 (a) **Identify three** of the component parts after a problem has been **decomposed**.

.....

.....

.....

(3 marks)

(b) **State three** sub-systems of a computer.

.....

.....

.....

(3 marks)

(c) **Describe** how the **use of sub-systems** can help in troubleshooting problems within a computer system.

.....

.....

(2 marks)

3 (a) Verification checks can be made on input data.

Tick (✓) one box to show which check is a verification check on input data.

- A.** checksum
- B.** double entry check
- C.** echo check
- D.** parity check

(1 mark)

(b) A validation check is used to make sure that any value that is input is an integer between 30 and 200 inclusive.

Give one example of each type of test data to check that the validation check is working as intended. Each example of test data must be different.

Give a reason for each of your choices of test data.

- **Normal test data**
- **Abnormal test data**
- **Extreme test data**

.....

.....

.....

.....

.....

.....

(6 marks)

- 4 (a) An algorithm has been written in pseudocode to input some numbers. It only outputs any numbers that are greater than or equal to 100. The number 999 is not output and stops the algorithm.

```
INPUT Number
WHILE Numbers <> 999 DO
  IF Number > 100
    THEN
      OUTPUT Number
    ENDIF
  ENDWHILE
OUTPUT Number
```

Identify the four errors in the pseudocode and suggest corrections.

(4 marks)

- (b) Write a pseudocode statement to change the corrected algorithm to output all numbers between 100 and 200 inclusive.

You **do not** need to rewrite the whole algorithm

(2 marks)

5 (a) Four programming concepts and four descriptions are shown.

Draw one line to connect each programming concept to the most appropriate description

Programming concept

Library routine

Structure diagram

Procedure

Function

Description

A subroutine that may **not** return a value.

A standard subroutine that is available for immediate use.

A subroutine that can be used in an assignment statement.

An overview of a program or subroutine.

(3 marks)

(b) Explain the purpose of the library routines **DIV** and **ROUND**.

(4 marks)

(c) **State two features** that should be included to create a maintainable program.

Give a reason why each feature should be used.

.....

.....

.....

.....

(4 marks)

- 6 (a) A database table called **TVRange** shows the main features and prices of a range of televisions.

TVCode	ScreenSize	Satellite	SmartTV	SoundBar	Price\$
TV90SaSmSd	90	YES	YES	YES	9750.00
TV75SaSmSd	75	YES	YES	YES	8500.00
TV75SaSd	75	YES	NO	YES	8000.00
TV65SaSmSd	65	YES	YES	YES	6000.00
TV65SmSd	65	NO	YES	YES	5000.00
TV65SaSd	65	YES	NO	YES	5000.00
TV55SaSmSd	55	YES	YES	YES	4000.00
TV55SaSd	55	YES	NO	YES	3500.00
TV55SmSd	55	NO	YES	YES	3500.00
TV50SaSmSd	50	YES	YES	YES	2500.00
TV50Sa	50	YES	NO	NO	1750.00
TV50Sm	50	NO	YES	NO	1750.00
TV40Sa	40	YES	NO	NO	1200.00
TV40	40	NO	NO	NO	950.00
TV32	32	NO	NO	NO	650.00

Give the name of the field that is most suitable to be the primary key.

State the reason for this choice.

(2 marks)

- (b) The database uses the data types:

- text
- character
- Boolean
- integer

- **real**
- **date/time**

Complete the table to show the most appropriate data type for each field. Each data type must be different.

Field	Data type
TVCode	
ScreenSize	
SmartTV	
Price\$	

(2 marks)

7 A database table called TVRange shows the main features and prices of a range of

televisions.

TVCode	ScreenSize	Satellite	SmartTV	SoundBar	Price\$
TV90SaSmSd	90	YES	YES	YES	9750.00
TV75SaSmSd	75	YES	YES	YES	8500.00
TV75SaSd	75	YES	NO	YES	8000.00
TV65SaSmSd	65	YES	YES	YES	6000.00
TV65SmSd	65	NO	YES	YES	5000.00
TV65SaSd	65	YES	NO	YES	5000.00
TV55SaSmSd	55	YES	YES	YES	4000.00
TV55SaSd	55	YES	NO	YES	3500.00
TV55SmSd	55	NO	YES	YES	3500.00
TV50SaSmSd	50	YES	YES	YES	2500.00
TV50Sa	50	YES	NO	NO	1750.00
TV50Sm	50	NO	YES	NO	1750.00
TV40Sa	40	YES	NO	NO	1200.00
TV40	40	NO	NO	NO	950.00
TV32	32	NO	NO	NO	650.00

Complete the structured query language (SQL) query to return the television (TV) code, screen size and price of all Smart TVs in the database table

SELECT TVCode,.....,

.....TVRange

WHERE SmartTV =.....;

.....

.....

.....

.....

(4 marks)

8 (a) Four logic gates and **five** standard symbols for logic gates are shown

Draw one line to link each logic gate to its standard symbol. Not all standard symbols will be used.

Logic gate	Standard symbol
AND	
OR	
NAND	
NOT	
	

(4 marks)

(b) Consider this logic expression.

$$X = (A \text{ AND } B) \text{ OR } (B \text{ AND NOT } C)$$

Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify this logic expression.



(4 marks)

(c) Consider the following logic statement:

$$X = ((A \text{ OR } B) \text{ AND } (\text{NOT } (B \text{ XOR } C)) \text{ AND } C)$$

Draw a logic circuit to represent the given logic statement.

Do not attempt to simplify the logic statement. All logic gates must have a maximum of two inputs.



(5 marks)

- 9 A one-dimensional (1D) array `Days[]` contains the **names of the days of the week**. A two-dimensional (2D) array `Readings[]` is used to **store 24 temperature readings**, taken once an hour, for each of the seven days of the week. A 1D array `AverageTemp[]` is used to **store the average temperature** for each day of the week.

The position of any day's data is the same in all three arrays. For example, if Wednesday is in index 4 of `Days[]`, Wednesday's temperature readings are in index 4 of `Readings[]` and Wednesday's average temperature is in index 4 of `AverageTemp[]`

The temperature readings are in **Celsius** to **one decimal place**. Temperatures can only be from **-20.0°C to +50.0°C inclusive**.

Write a program that meets the following requirements:

- input and validate the hourly temperatures for one week
- calculate and store the average temperature for each day of the week
- calculate the average temperature for the whole week
- convert all the average temperatures from Celsius to Fahrenheit by using the formula $\text{Fahrenheit} = \text{Celsius} * 9/5 + 32$
- output the average temperature in Celsius and in Fahrenheit for each day
- output the overall average temperature in Celsius and in Fahrenheit for the whole week.

You must use **pseudocode** or **program code** and **add comments to explain** how your code works.

You do **not** need to declare any arrays, variables or constants; you may assume that this has already been done.

All inputs and outputs **must** contain suitable messages.

All data output **must** be rounded to one decimal place

You will need to **initialise** and **populate** the array `Days[]` at the start of the program

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(15 marks)